

OPERATING SYSTEMS

HW 2 Code Document: Producer-Consumer

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This assignment is related with how thread works and how to manage it via a classic problem like it's the producer-consumer problem. Given a file, the producers have to read it, save the information and keep it until the consumers want to read that information. The key thing here is to manage the time to prohibit that a consumer to read when there's no information, a producer to produce when a consumer is reading the information etc.

We'll be able to do this with `pthread_create`, `pthread_mutex_init`, `pthread_join` or `pthread_exit` to manage the threads the way we want plus the use of flags to decide in which conditions the consumer or the producer can join and use the resources.

The code has some additional information such as the consumer lines read and the content of the lines, the quantities of each word read, the total lines and the elapsed time.

I have used Visual Studio Code with a wsl terminal

1: Compile the file with `gcc -o (final name) (original_name.extension)`

2: `./ (final name) (file_to_read.extension) (number of producers) (number of consumers)`

Here's an example with the given file in hw1 (helloworld.c):

```
saulgil@DESKTOP-3QQ48BQ:/mnt/c/Users/Usuario/Documents$ gcc -o prod_cons1 prod_cons1.c
^[[Asaulgil@DESKTOP-3QQ48BQ:/mnt/c/Users/Usuario/Documents$ ./prod_cons1 helloworld.c 2 8
main continuing
Cons_d1ffb6c0: [00:00] #include <stdio.h>
Cons_d1ffb6c0: [01:00]
Cons_e0ffb6c0: [00:01] int main(void){
Cons_e0ffb6c0: [01:02]
Cons_d37fe6c0: [00:03]     printf("Hello world\n");
Cons_e1ffd6c0: [00:04]
Cons_e1ffd6c0: [01:01]
Prod_e27fe6c0: 6 lines
Cons_e0ffb6c0: [02:05] }Prod_e2fff6c0: 2 lines
Cons_d27fc6c0: 0 lines
Cons_d2ffd6c0: 0 lines
Cons_e1ffd6c0: 2 lines
Cons_e17fc6c0: 0 lines
Cons_d37fe6c0: 1 lines
Cons_e0ffb6c0: 3 lines
Cons_d1ffb6c0: 2 lines
Cons_d3fff6c0: 0 lines
main: consumer_0 joined with 2
main: consumer_1 joined with 0
main: consumer_2 joined with 3
main: consumer_3 joined with 0
main: consumer_4 joined with 1
main: consumer_5 joined with 0
main: consumer_6 joined with 0
main: consumer_7 joined with 2
main: producer_0 joined with 2
main: producer_1 joined with 6
```

```
==== Character statistics (A-Z) ====
A: 1
B: 0
C: 1
D: 4
E: 2
F: 1
G: 0
H: 2
I: 6
J: 0
K: 0
L: 4
M: 1
N: 5
O: 4
P: 1
Q: 0
R: 2
S: 1
T: 3
U: 1
V: 1
W: 1
X: 0
Y: 0
Z: 0
Total letters counted: 41
Total lines produced: 8, consumed: 8
Elapsed time: 0.011023 seconds
saulgil@DESKTOP-3QQ48BQ:/mnt/c/Users/Usuario/Documents$
```

The program read the lines as it is written, not the final output of the file, so it's better to use a .txt but the example is as valid as with a .txt

No freedays used yet.